Short-Term InSAR Displacement and Surface Change Around the Salt Lake City Airport



Photo courtesy of Ardurra, from the Salt Lake City Airport Building Demolition project page.

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Abstract

This project uses Sentinel-1 SAR data to evaluate short term surface motion and longer term surface change around the Salt Lake City Airport and the northwest industrial corridor. A line-of-sight displacement product from October 22 to November 03 was processed, masked with coherence, and converted from meters to centimeters. The short term LOS signal was then compared with a 2022 to 2024 VV backscatter change layer to highlight real surface modification. A 500 meter UTM grid was used to average LOS values and reduce noise. The LOS results mostly reflect atmospheric phase delay, while the VV change clearly captures real construction. Overall, the analysis shows how different InSAR products can be combined to evaluate stability and development around major infrastructure.

I. Research Question

A. Primary Question

Can short term ground displacement or surface changes be detected around the Salt Lake City Airport and the surrounding industrial area, and where do these changes occur.

B. Broader Importance

Airports and large industrial zones are built on engineered ground that can shift over time. InSAR provides a way to check for movement across a wide area without installing instruments. Even if the short term motion is mostly atmospheric noise, the process still shows what areas are stable, what areas lose coherence, and what parts of the city are changing due to construction.

II. Description of Data Sets

A. Sentinel-1 SAR

- Sentinel-1A and 1B
- IW mode
- VV polarization
- Wavelength: 5.6 cm
- Spatial resolution: about 10 meters

B. Products Used

LOS Displacement (HyP3)

- Oct 22 to Nov 03
- Delivered in meters
- Converted to centimeters
- Final range: about 1.17 to 2.90 cm

Coherence

- Same dates
- Used to remove low quality pixels
- Threshold: 0.3

VV Backscatter Change

- 2022 to 2024
- Highlights actual surface modification like new pavement and construction

500 Meter Grid

- Used for zonal statistics

OSM Roads and Buildings

- For visual context

III. Analysis Techniques

A. Methods used

- LOS displacement conversion and masking
- VV backscatter differencing
- Grid based LOS summarization
- Visual comparison with OSM data

B. Why these methods

LOS displacement captures short term phase changes that might indicate surface motion, even if most of it turns out to be atmospheric. Coherence is needed to filter out unstable areas so the LOS values come from reliable pixels. The VV backscatter change shows real long term modification on the ground, like new pavement or construction. The grid based approach smooths out pixel level noise and makes it easier to compare different parts of the scene.

C. Steps Followed

1. Imported the HyP3 LOS raster and converted it from meters to centimeters.
2. Imported coherence and masked out values under 0.3.
3. Calculated VV backscatter change between 2022 and 2024.
4. Created a 500 meter UTM grid and calculated mean LOS values for each cell.
5. Subtracted the global average to create LOS anomalies.
6. Overlaid OSM buildings and roads for context.

IV. Results

A. Coherence

The coherence map defines where the InSAR result is trustworthy. Pavement, runways, and rooftops hold coherence well, so the interferogram stays stable in those areas. Coherence drops off fast in soil yards, water edges, gravel lots, and anything disturbed or vegetated. These low-coherence zones explain why the LOS map ends up with so many masked areas. Only the engineered surfaces around the airport consistently support a clean LOS signal.

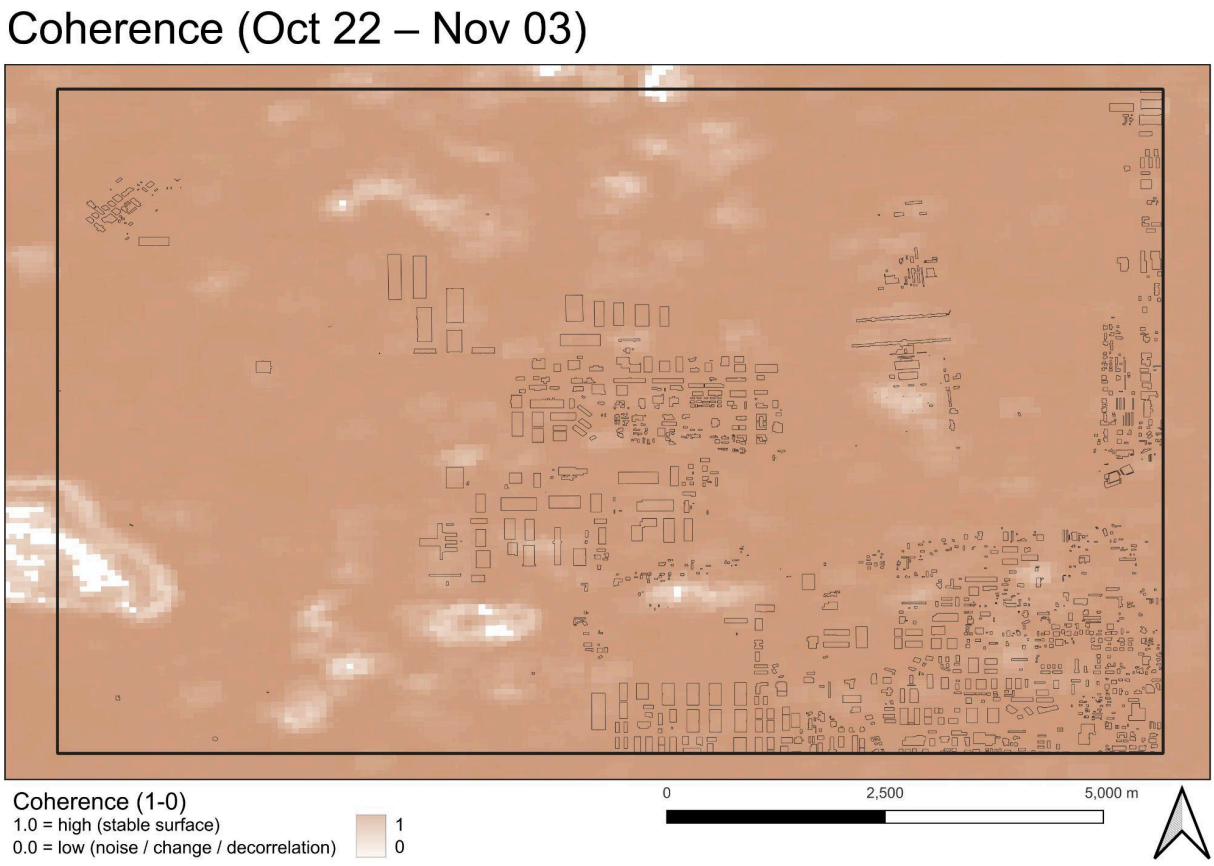


Figure: Coherence (Oct 22 to Nov 03)

Coherence values from 0 to 1. High coherence corresponds to stable artificial surfaces. Low coherence appears in areas with vegetation, water, or recent disturbance.

B. LOS Displacement

The LOS values range from about 1.17 to 2.90 cm, but none of that is real deformation. The interferogram is only 12 days long, so the map is dominated by atmospheric phase delay and a broad ramp across the scene. After applying the coherence mask, only scattered high-coherence pixels remain, and the variation across those pixels is just atmospheric noise. No part of the study area shows displacement large enough or patterned enough to indicate true ground motion.

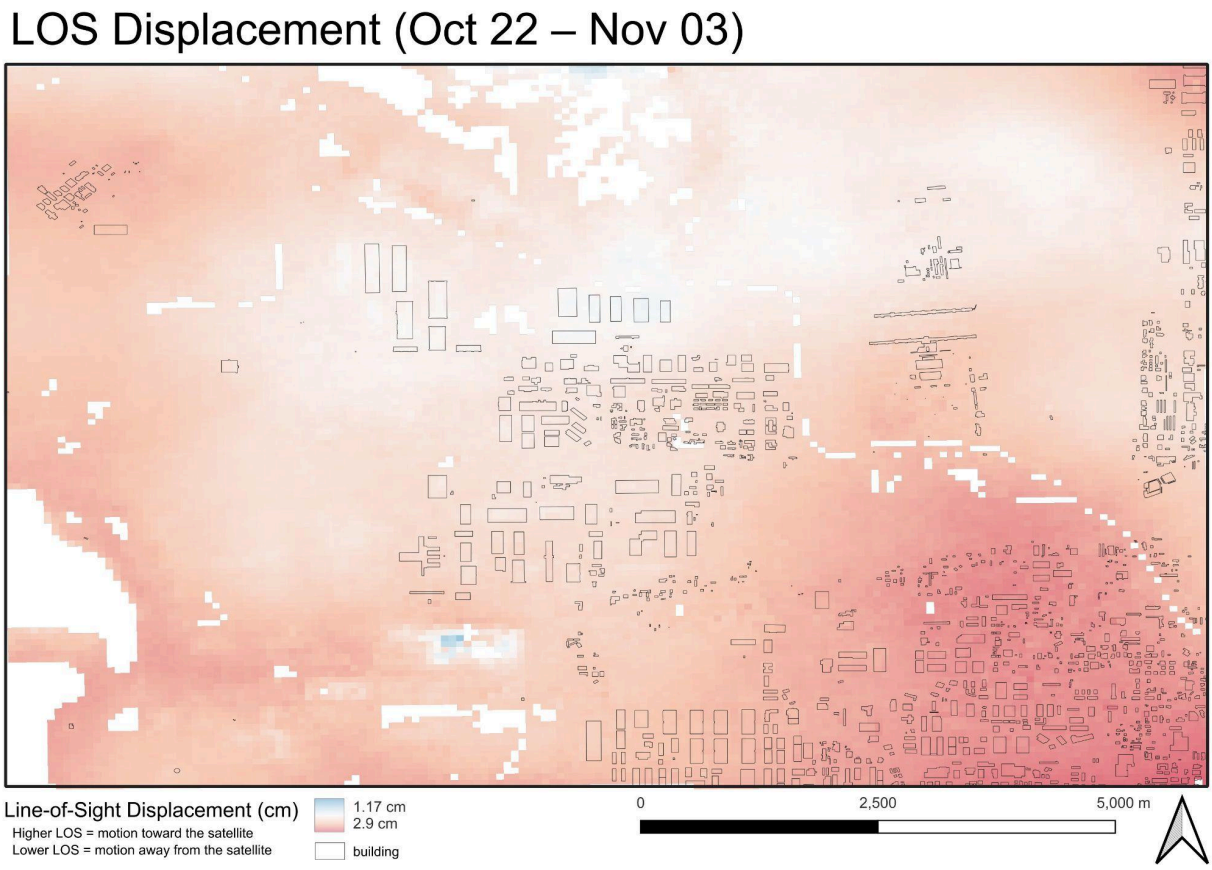


Figure: LOS Displacement (Oct 22 to Nov 03)

Values in centimeters. Higher values mean motion toward the satellite. Lower values mean motion away. Most variation is atmospheric and not actual ground movement.

C. VV Backscatter Change

The VV backscatter change from 2022 to 2024 captures actual physical modification on the ground. Red areas indicate stronger backscatter (new pavement, building pads, resurfacing). Blue areas mark reduced backscatter (grading, demolition, bare soil). These patterns line up with known construction zones around the airport and along the 5600 West industrial corridor. Unlike LOS, this map reflects real change, not atmosphere.

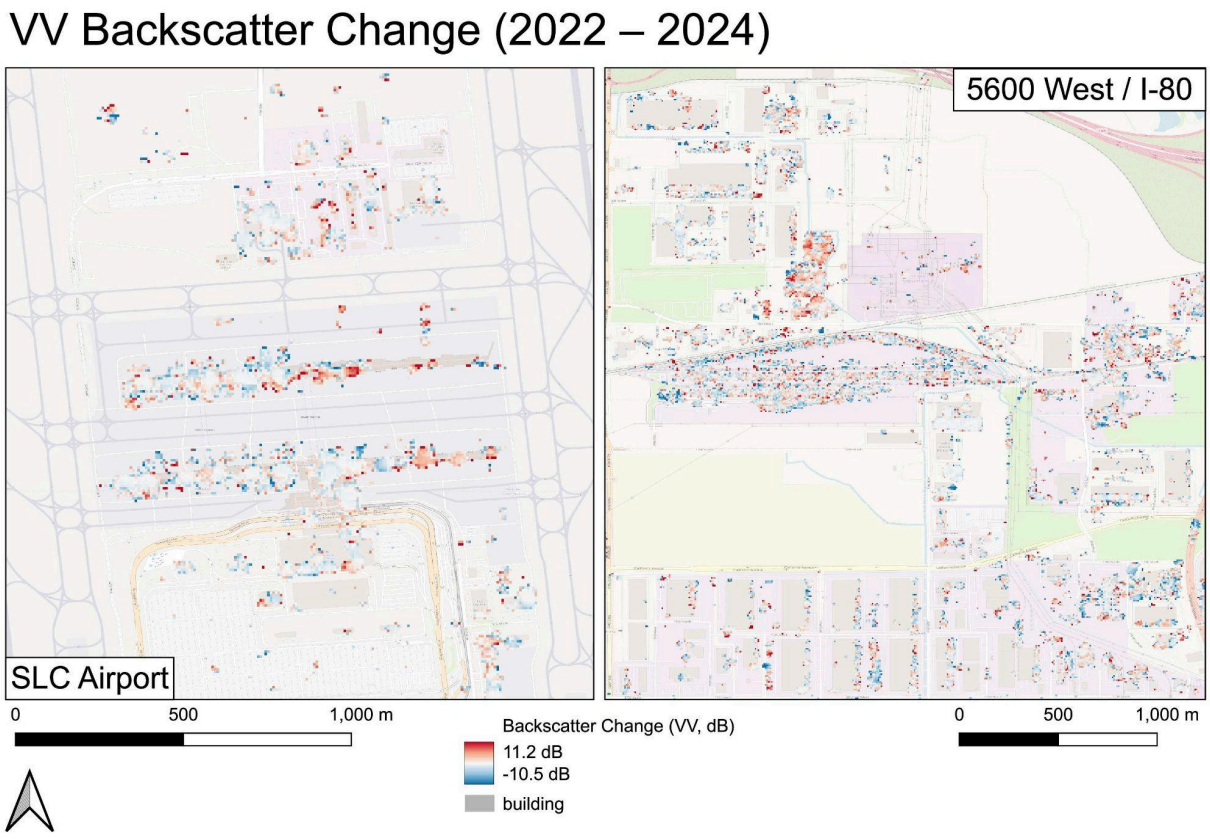


Figure: VV Backscatter Change (2022 to 2024)

Differences in VV backscatter reveal real physical modification over the two year period.

D. Grid Based LOS Displacement

The 500-meter grid smooths out pixel-level noise and makes the LOS patterns easier to read. The anomalies fall within ±0.5 cm, which is consistent with atmospheric variation and not physical deformation. No grid cell shows a coherent displacement pattern that would indicate subsidence or uplift. The grid result reinforces the conclusion that the 12-day LOS signal contains no measurable ground motion.

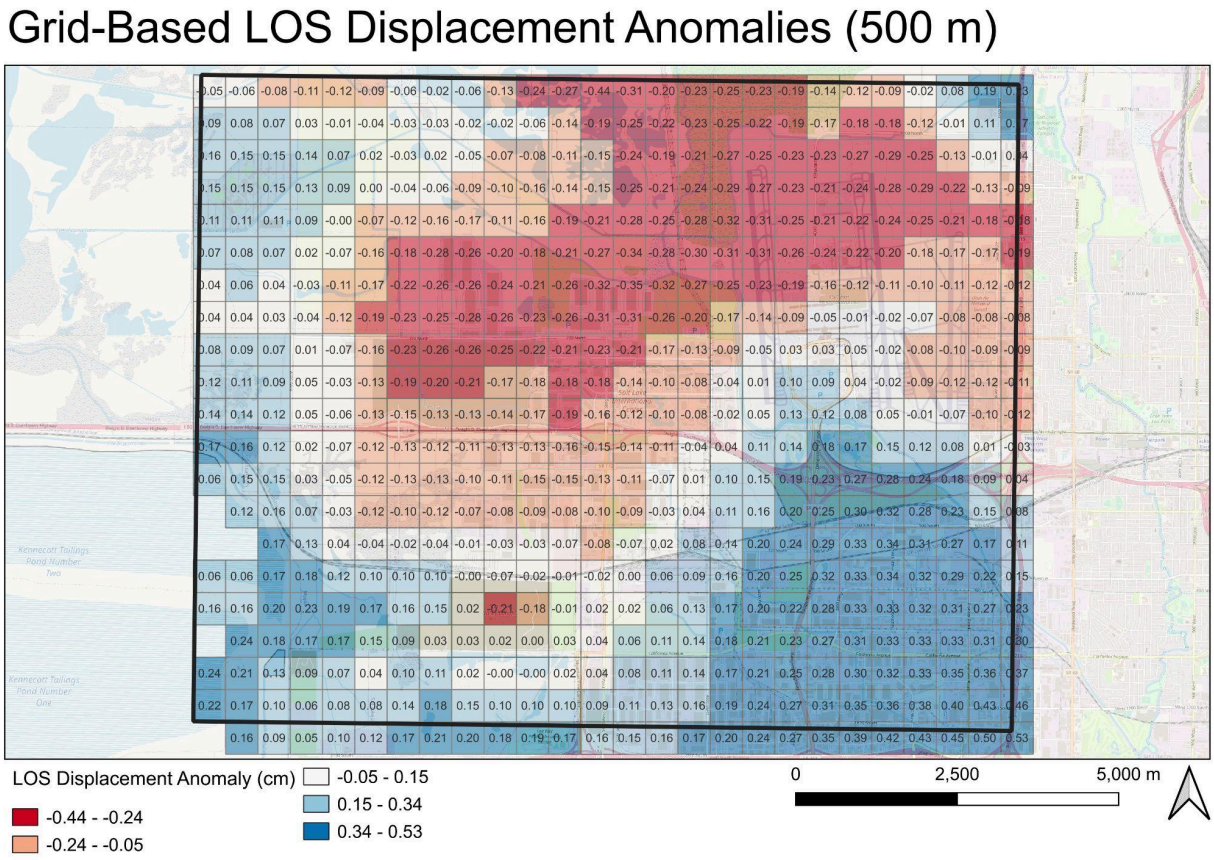


Figure: Grid-Based LOS Displacement Anomalies (500 m)

Mean LOS displacement per grid cell. Values reflect atmospheric variation, not deformation.

V. Discussion

A. Interpretation

The LOS pattern is mainly atmospheric because the interferogram spans only 12 days. Coherence is low in several areas, especially near the lake and disturbed soil, which limits LOS accuracy. The VV backscatter change map gives the clearest signal of real surface modification. The grid based LOS results support the conclusion that no measurable ground deformation is present.

B. Problems Encountered

- Large areas of low coherence removed most of the LOS pixels.
- The short time span created strong atmospheric ramps.
- VV change can identify construction but cannot confirm vertical movement.
- PDF export required specific settings to avoid visual seams.

VI. Conclusions

A. Can the Research Question Be Answered?

Yes. The combination of LOS displacement, coherence, VV backscatter change, and grid based summarization gives a clear picture of what was happening in the study area. The short term LOS values do not show any sign of real deformation, and the VV backscatter change pinpoints where actual surface modification occurred.

B. Final Answer

There is no meaningful ground movement in the study area during the 12 day interferogram. The LOS displacement field is dominated by atmospheric effects, which is expected for such a short time span. The VV backscatter change map is the main source of real surface information, and it highlights construction and redevelopment between 2022 and 2024 around the airport and the northwest industrial corridor. Overall, the area appears stable during the study window.

C. Improvements

A longer time series of interferograms would allow real deformation to stand out from atmospheric noise and would create a more reliable trend. Adding GNSS or LiDAR would help validate any long term motion. Using multiple baselines and more dates would also reduce the influence of low coherence and provide a clearer understanding of surface stability across the region.